温州肯恩大学
WENZHOU-KEAN UNIVERSITY

GMBA 5885 W01
Strategic Mgt Global Info Sys

# Introduction to MIS

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# Learning Objectives

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By the end of this session, you should be able to:

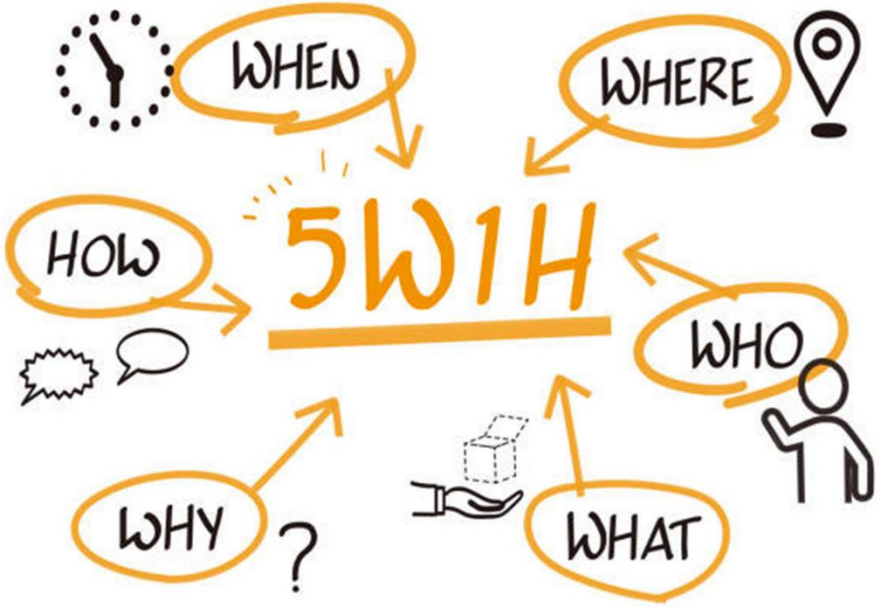
- Explain how information systems (IS) transform busine
- Identify key strategic objectives supported by IS
- Define an IS and describe how it works
- Understand the organizational, managerial, and technological dimensions of IS
- Recognize emerging technologies shaping modern IS
- Evaluate opportunities and risks associated with IS

# Outline

Today's Structure

- Why IS matters (business transformation)
- What is an IS
- How IS works
- Where IS creates value (strategy)
- What technologies are driving change
- What challenges managers face
- Application: UPS case

5W1H Logic

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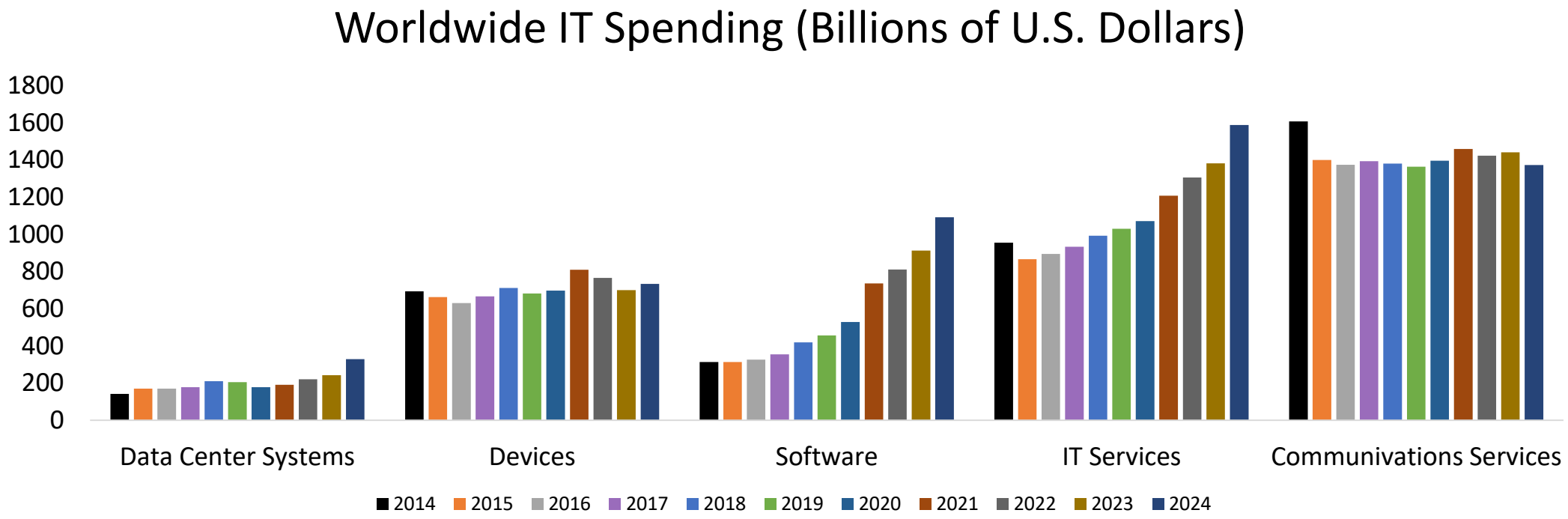
# IS Transforms Business




*Source: Gartner*

*"If a business isn't using information systems connected to the Internet and mobile apps, chances are it isn't being as effective as it could be."*

*Laudon 18th Ch1*

# IS Transforms Business

- Data is a critical production factor (like capital and labor)
- IS converts data → insights → decisions → value
- Digital technologies reshape:
  - Business models (e.g., platforms, ecosystems)
  - Work and skills (AI, automation)
  - Competitive dynamics

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> *"Every global company is becoming a tech company … we're seeing technology as a critical component for business success."*
>
> *—— Jeremy King, CTO@Walmart Labs*

*Pearlson 8th Introduction*

# Why Managers Must Understand IS

**IS decisions** ↔ business decisions

Managers must:

- Align IS with business strategy
- Evaluate technology investments
- Manage digital transformation
- Mitigate risks (cybersecurity, AI bias, system failure)

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👉 Key insight: You don’t need to code — but you must **think digitally**

# Skills for Successful IT Use by Managerial Role

| Managerial Role | Skills |
|---|---|
| Visionary | Creativity, curiosity, confidence, focus on business solutions, flexibility |
| Informational and Interpersonal | Communication, listening, information gathering, interpersonal skills |
| Structural | Project management, analytical, organizational, planning, leading, controlling |

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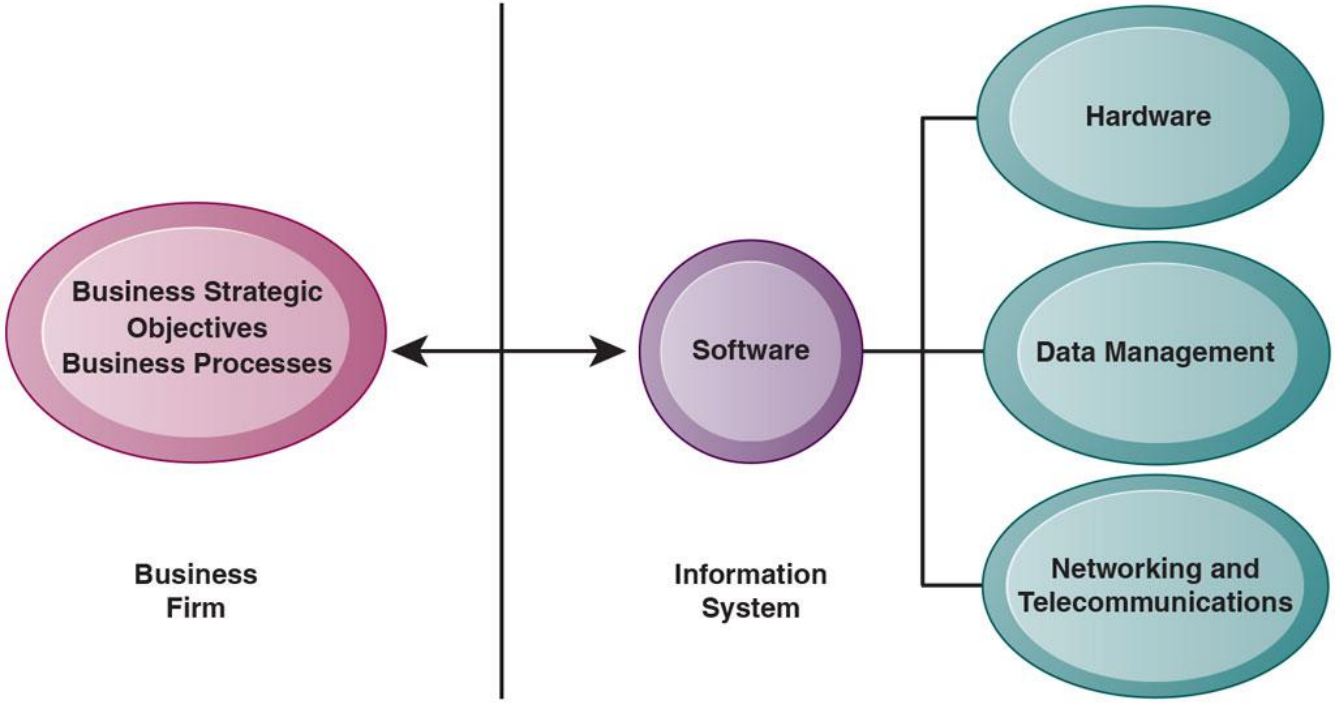
# Strategic Role of IS

Strategy ↔ Information Systems are Interdependent

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Understanding IS = understanding how strategy is executed

# Six Strategic Business Objectives Enabled by IS

1. Operational excellence
   - Improve efficiency, productivity, and cost control
   - E.g., Walmart's real-time supply chain systems
2. New products, services, and business models
   - Enable digital innovation and new ways of creating value
   - E.g., Apple iTunes transforming music distribution
3. Customer and supplier intimacy
   - Deep understanding → personalization and stronger relationships
   - E.g., Ritz-Carlton (customer experience)
4. Improved decision making
   - Real-time, data-driven decisions (vs. intuition)
   - E.g., Coca-Cola dashboards for operations and sales
5. Competitive advantage
   - Doing things better, faster, or cheaper than competitors
   - Result of achieving the above objective
6. Survival
   - IS becomes a necessity due to:
     - Industry standards (e.g., ATMs in banking)
     - Regulatory requirements (e.g., data retention)

# Quiz1

Which of the following is **not** an important business objective driving investment in information systems and technologies?

A. Operational excellence

B. Improved decision making

C. Improved employee benefits

D. New products, services, and business models

# What Is an IS/MIS

- Information Technology (IT)
  - Hardware + software
  - Technology alone does not create value
- Management Information System (MIS/IS)
  - A set of interrelated components that collect (or retrieve), process, store, and distribute **information** to
    - Support decision making, coordination and control
    - Help analyze problems, visualize complex issues, and develop new products or services
  - IT + people + processes

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# What Is an IS

| | Data → | Information → | Knowledge |
|---|---|---|---|
| **Definition** | Raw facts about the world | Processed data with meaning and relevance | Interpreted information with context and judgement |
| **Key Features** | • Structured and easily captured<br>• Quantified<br>• Easy to store and transfer | • Processed and organized<br>• Require context and analysis<br>• Supports understanding | • Context-rich and experience-based<br>• Often tacit<br>• Difficult to codify and transfer |
| **Example** | Daily inventory report of all items | Daily inventory report of items with low levels of stock | Report of which items need to be reordered after considering inventory levels, anticipated labor strikes, and a flood affecting a supplier |

*Pearlson 8th Introduction*

# What Is an IS

Different managerial roles require different types of information — IS must be designed accordingly.

| | Top Management | Middle Management | Operational / Supervisory |
|---|---|---|---|
| **Time Horizon** | Long-term (years) | Medium-term (weeks, months, years) | Short-term (daily) |
| **Information Detail** | Highly aggregated<br>Less precise<br>Future-oriented | Summarized<br>Integrated<br>Often financial | Highly detailed<br>Highly accurate<br>Often nonfinancial |
| **Source** | Primarily external | Primarily internal with limited external | Primarily Internal |
| **Decision Type** | Strategic<br>High judgment & creativity | Tactical<br>Analytical | Operational<br>Rule-based |

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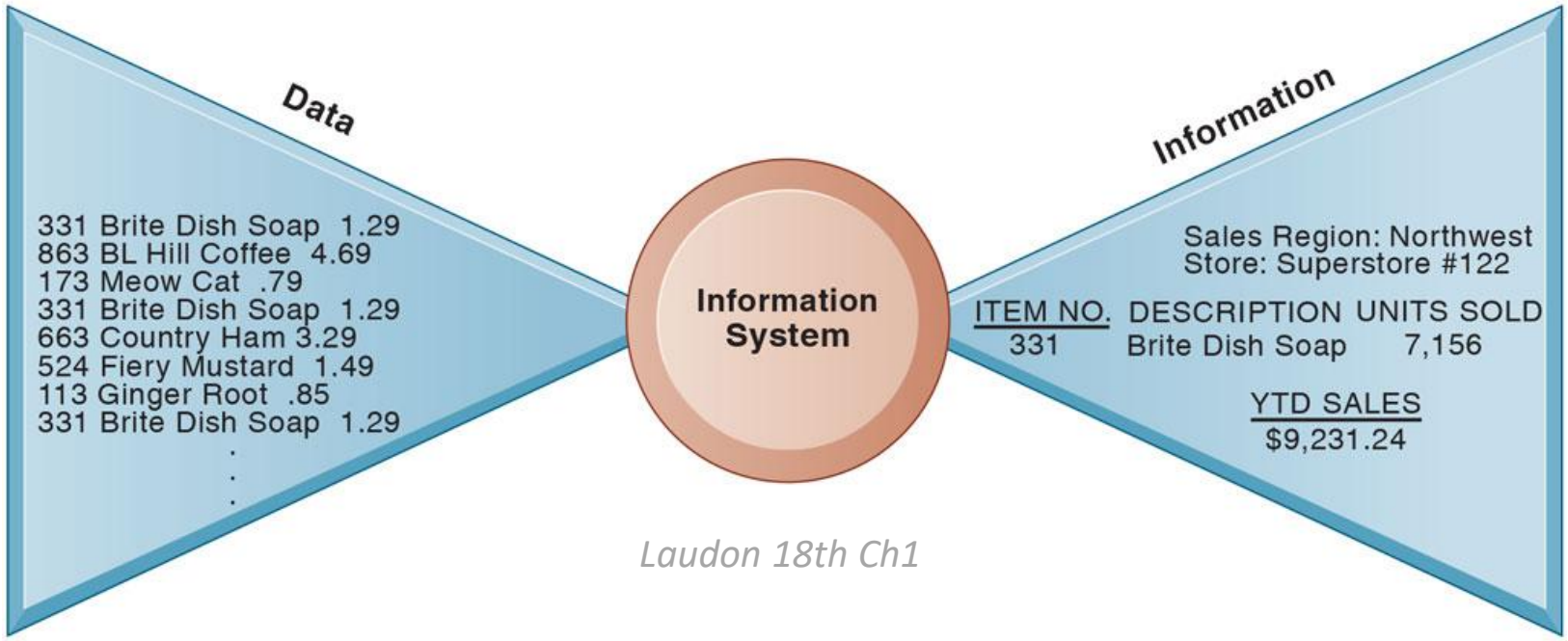
# What Is an IS

**Example:** Raw data from a *supermarket checkout counter* can be processed and organized to produce meaningful information.

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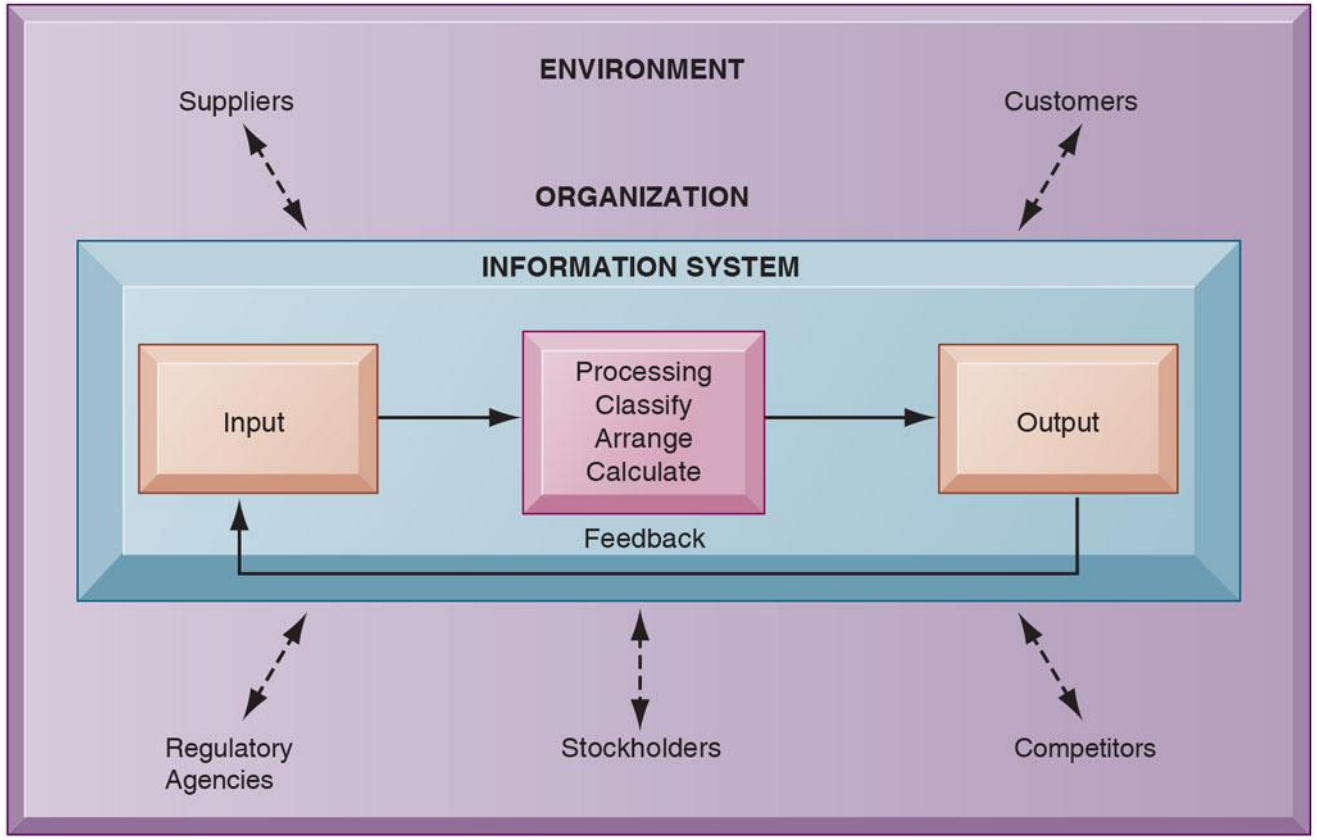
Streams of raw facts

Data shaped into meaningful, useful form

# How an IS Works

## Functions of an IS

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# How an IS Works

- Three Core Activities
  - **Input:** Captures raw data from inside or outside the organization
  - **Processing:** Converts data into meaningful information
  - **Output:** Distributes processed information to users and decision-makers
- Feedback
  - Output returned to relevant people or systems to improve future inputs and processes
- External Actors
  - Customers, suppliers, competitors, stockholders, and regulators continuously interact with the system

# Quiz2

**Processing** involves which of the following?

A. Returning output to appropriate members of the organization to help them evaluate or correct the input stage.

B. Transferring information to the people who will use it or to the activities for which it will be used.

C. Capturing or collecting raw data from within the organization or from its external environment.

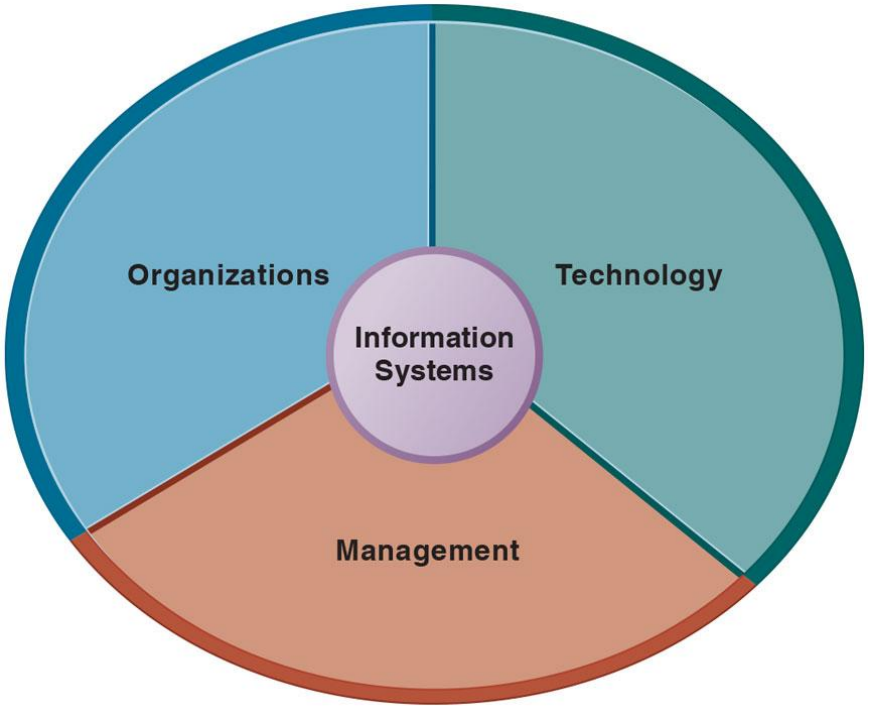
D. Converting data into a meaningful form by classifying, arranging, and performing calculations on it.

# IS as a Sociotechnical System

IS = Technology + Organization + People

☞ Performance depends on alignment across:

- Organizational structure and culture
- Management decisions and incentives
- Technology infrastructure

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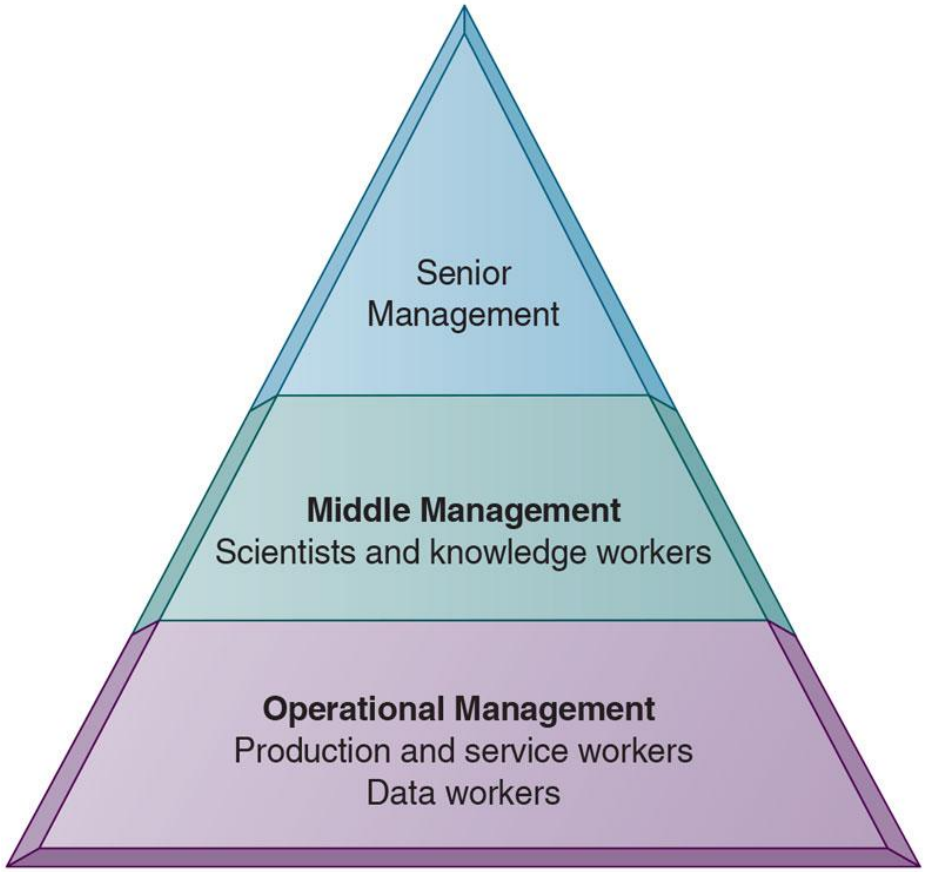
# IS Dimension 1: Organization

IS and organizations are mutually shaping

- How IS supports the organization
  - Enables core functions: marketing, finance, HR
  - Integrates key business processes (e.g., ordering, hiring, product development)
  - Embeds and automates organizational rules and workflows
- How organization shapes IS
  - Culture influences adoption and use
  - Structure & processes determine system design
  - Politics & interests affect implementation decisions

Levels in a Firm

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# IS Dimension 2: Management

Managers use IS to make decisions, execute strategy, and drive innovation

- Role of Management
  - Interpret business challenges and set strategic direction
  - Allocate resources and coordinate people, processes, and technology
  - Balance control, planning, and innovation
- How IS Enables Management
  - Provide timely, accurate, and relevant information
  - Support analysis, forecasting, and decision-making
  - Enable insight generation and problem solving

# IS Dimension 3: Technology

Technology provides the foundation that enables information systems

- Core Components
  - **Hardware:** devices for input, processing, storage, and output
  - **Software:** applications and systems that control operations
  - **Data management:** tools to store, organize, and access data
  - **Networking & telecommunications:** enable connectivity and data exchange
- IT Infrastructure
  - Integrated set of technologies and supporting capabilities
  - Forms the platform on which IS are built and scaled

# Quiz3

Which of the following makes **long-range** strategic decisions in a firm?

A. Senior management

B. Knowledge workers

C. Middle management

D. Operational management

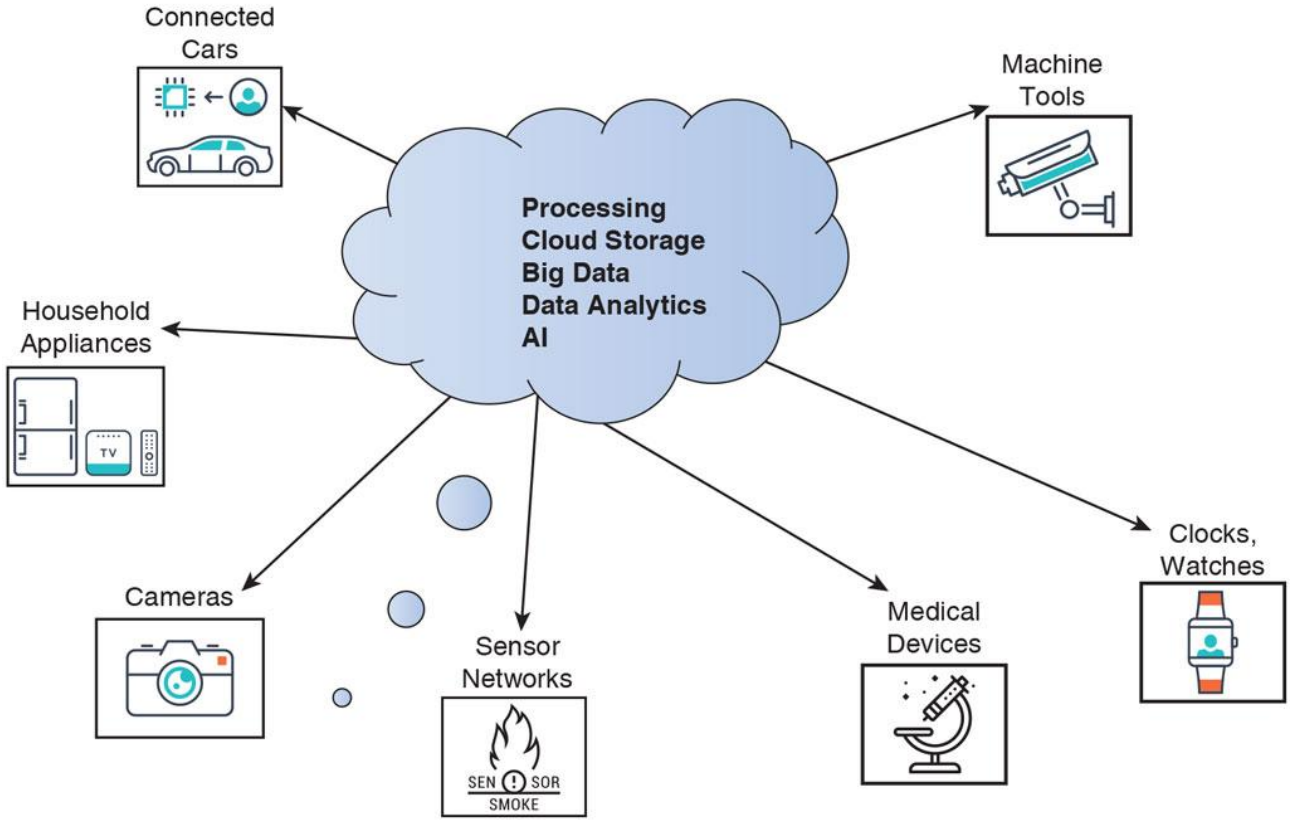
# Key Technologies Shaping Modern IS

Modern IS powered by an integrated digital stack that transforms data into business value

- Internet of Things (IoT)
  - Network of connected devices & sensors
  - Generates real-time data for monitoring and automation
  - Primary source of big data
- Big Data
  - High volume, velocity, and variety of data
  - Enables advanced analytics, forecasting, and optimization
- Cloud Computing
  - On-demand access to computing, storage, and software
  - Provides scalable infrastructure for data and AI
- Artificial Intelligence (AI)
  - Systems that simulate human intelligence (prediction, decision-making)
  - Includes ML, DL, GenAI
  - Drives automation and innovation

# Key Technologies Shaping Modern IS

IoT → Data Generation → Big Data → Cloud Processing → AI Insights → Business Decisions



# Complementary Assets

- Why IT alone is not enough
  - Firms with similar IT investments often achieve very different outcomes
  - Value is created when technology is effectively integrated into the organization
- How IS creates value
  - Improves productivity and efficiency
  - Enhances decision-making and business processes
  - Strengthens revenue growth and market position
- Required complementary assets
  - Organizational: culture, structure, business processes
  - Managerial: leadership, skills, incentives
  - Social: infrastructure, standards, regulation

Sustainable advantage comes from aligning IT with organizational and managerial capabilities

# Quiz4

Which of the following is **not** an example of a **complementary social asset** required to optimize returns from information technology investments?

A. Government and private sector standards

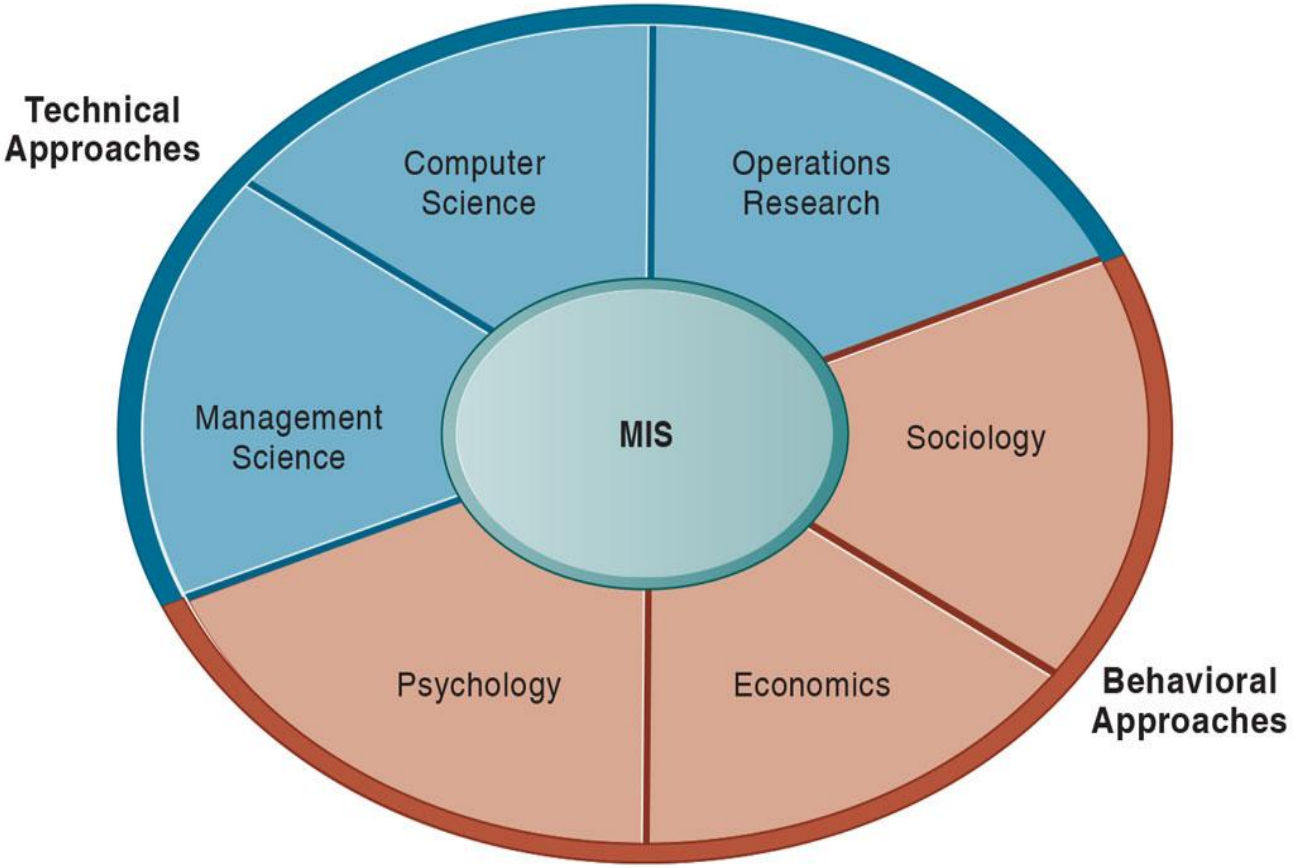
B. IT-enriched educational programs

C. Laws and regulations creating fair, stable market environments

D. Efficient business processes

# MIS as a Research Field

- Multidisciplinary (technology + behavior)
- Focus: design, use, and impact of IS
- Combines:
  - Technical perspective
  - Behavioral/organizational perspective

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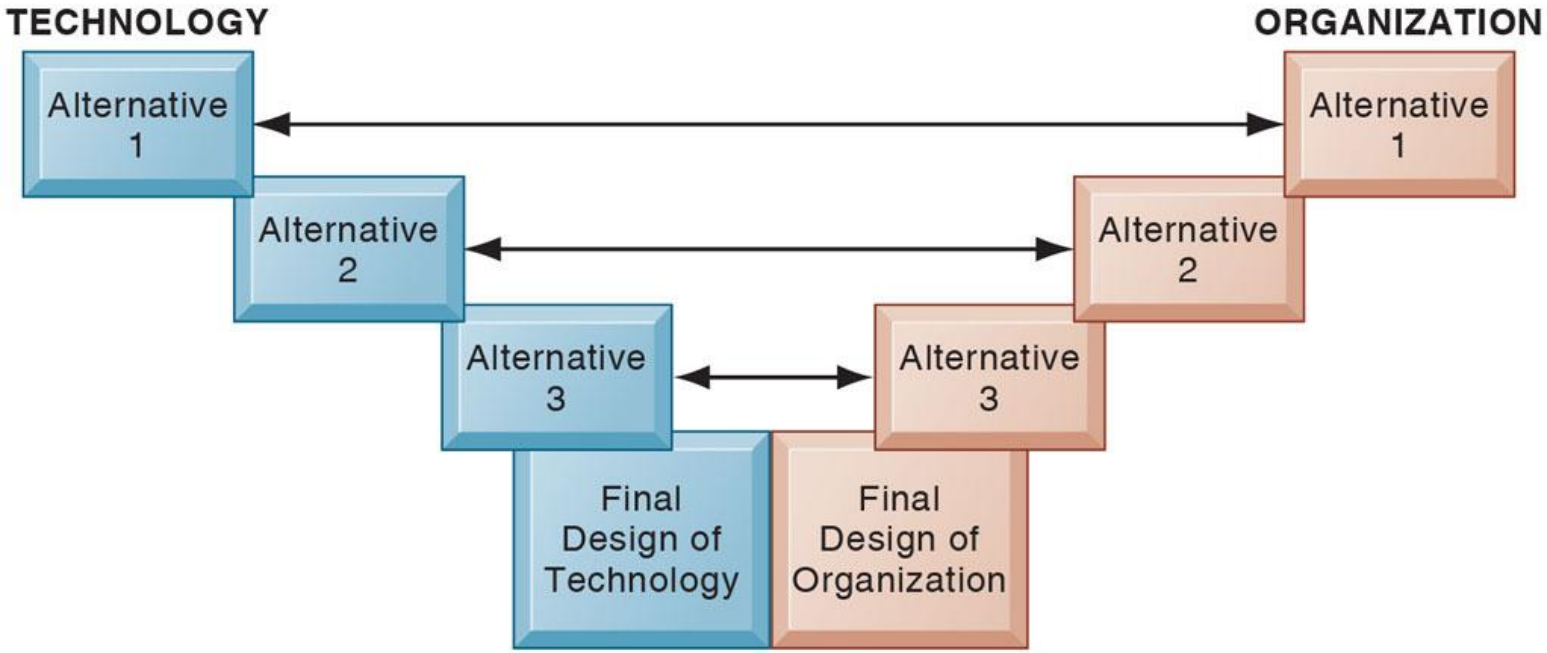
# MIS as a Research Field

Research Topic in MIS (MIS Quarterly Research Curations)

- Information Systems Development
- IS Sourcing
- IT Project Management
- Data Management
- Information Systems Alignment
- IS Use
- Knowledge Management
- Health Information Technology
- IT-supported Collaboration
- Online Word-of-Mouth
- Platforms and Ecosystems
- IT Workforce
- Information Privacy
- IS Control & Governance
- Societal Implications of ICTs

# A Sociotechnical Perspective on MIS

Optimal organizational performance is achieved by jointly optimizing both the **social** and **technical** systems used in production.

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# Key Challenges of Modern MIS

| Technology Changes | Business Challenges |
|---|---|
| **Cloud Computing**<br>Organizations move IT operations to cloud-based platforms for scalability and cost efficiency. | How much IT infrastructure should move to the cloud? What are the trade-offs in cost, control, and security? |
| **Big Data & IoT**<br>Firms collect vast data from sensors, web traffic, and social media to improve decisions. | Does your firm have the tools and talent to analyze and leverage big data effectively? |
| **Artificial Intelligence (AI)**<br>Systems perform tasks requiring human-like intelligence—recognizing patterns, predicting, learning. | Where can your company apply AI? What benefits, costs, and expertise are needed for implementation? |
| **Mobile Platforms**<br>Smartphones, tablets, and wearables enable collaboration and connectivity anywhere. | Is your firm maximizing mobile capabilities for employees and customers? How can it enhance value? |

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# Key Challenges of Modern MIS

| Management | Business Challenges |
| --- | --- |
| **Return on Investment (ROI)**<br>Firms invest heavily in information systems but often struggle to measure the value gained. | How can your firm accurately assess the benefits of IT investments and identify more cost-effective alternatives? |
| **Online Collaboration & Social Networks**<br>Platforms like Google Workspace, Microsoft 365, Slack, and Zoom reshape teamwork and communication. | Is your firm effectively using digital tools to improve coordination and knowledge sharing? Which platforms add the most value? |

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# Key Challenges of Modern MIS

| Organizations | Business Challenges |
|---|---|
| **Security & Privacy**<br>Data protection and privacy have become critical public and regulatory concerns. | How secure is your company's data? Are its privacy policies adequate and compliant with evolving laws? |
| **Social Business**<br>Firms leverage social platforms (Facebook, X, TikTok, internal networks) to engage employees and customers. | Is your firm gaining measurable value from social business tools? How could engagement be deepened? |
| **Remote & Hybrid Work**<br>Cloud and mobile technologies enable flexible work models across locations. | What policies and technologies will shape your firm's future of work? What challenges and opportunities exist? |

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# Summary

- IS enables data-driven strategy and competitive advantage
- Supports six objectives: efficiency, innovation, intimacy, decisions, advantage, survival
- IS = collect → process → store → distribute information
- Value requires alignment: organization + management + technology
- Technologies (IoT, Big Data, Cloud, AI) are reshaping competition
- IT + complementary assets = business value